Examiner reports

Q2.

Overall, this question was answered well and was a good source of marks for most students. Over half the entry scored 10 or more of the 11 marks available.

Part (a). The vast majority of students answered correctly, showing confidence in finding partial fractions. The most common method was to substitute x = 3 and x = $-\frac{1}{3}$, although some students used simultaneous equations and others hybrid methods. Few mistakes were seen, although some students did manage to swap over the given coefficients A and B.

Part (b)(i). Many students approached this by writing out the full expression using their values of A and B and using the binomial expansions of both terms clearly and correctly to arrive at the correct answer. The common errors were not to take the 3 out of (3 – x) correctly, if at all, and / or to get confused with the index -1, this sometimes being combined with the 2 (value of A) to become $\frac{3}{2}$, 6 or $\frac{1}{6}$. The other common error was to drop the minus sign in the expansion of $\left(1-\frac{x}{3}\right)^{-1}$. Although most students made a clear attempt to combine the two expansions, some made sign errors. Very few students attempted to use the binomial theorem result from the formula book, or did not use the partial fractions and multiplied their expansions together with (7x - 1), although a few correct solutions were seen using this method.

Part (b)(ii). Most students were aware that this question involved the range of validity of the expansions, with many finding the range correctly but not relating this to the given value of 0.4; this didn't answer the question. Other students seemed to just make a guess, such as '*not enough terms have been considered*' or even '*because 0.4 is a decimal number*'.

Q3.

After a successful start with the partial fractions many students went awry in later parts of the question, although fully correct solutions were seen from about 16% of the entry.

(a) (i) The partial fractions were found correctly by the vast majority of students. Virtually all students knew what to do and had a correct opening line and only the occasional arithmetic error followed in the finding of A and B. The most popular method was to substitute x = –2 and x = $\frac{1}{3}$, although some used simultaneous equations and others a mix of the two methods, often choosing to substitute x = 0.

(ii) Again the vast majority of students had the correct form of the integrals but many errors were made in the coefficients, particularly with the (1 – 3x) term, often given simply as ln (1 – 3x) or + $\frac{1}{3}$ ln (1 – 3x) and the (2 + x) term was sometimes integrated $\frac{3}{2}$ ln(2 + x). Most students went on to evaluate their integrals in the correct way, although it was not always clear how they used the given limits and many made a sign error to obtain 3 ln2 – $\frac{1}{3}$ ln4, although ln1 = 0 was well known. Many students who had the definite integral correct could not convert to the form of the requested answer. Several of those who got as far as 3 ln2 + $\frac{2}{3}$ ln 2 did not see that all they needed to do was add 3 to $\frac{2}{3}$; the answer ln8 was common. The term rational may not have been sufficiently well known.

(b) (i) It was expected that students would require little work to find the value of C; it is immediate from the algebraic fraction by inspection from $-6x^2$ / $-3x^2$. However, relatively few noticed this. Many chose long division and found the correct value, but some used inefficient and time consuming methods based on undetermined coefficients. Some found C to a function of x although most gave C as a constant. Some students did not attempt this part of the question.

(ii) All those who attempted this part knew that the requested area was given by integrating the function from part (b)(i), and most students recognised that this linked the parts of the question together. They gained credit if they had an exact answer to part (a)(ii) and added 2 to it, although many had a sign error when applying the limits and subtracted 2. Some students did not see the connection and tried to do the integral again, this time often going awry with answers such as ln $(2 - 5x - 3x^2)$.

Q4.

In part (a), virtually all candidates showed confidence with finding the partial fractions and, bar a few slips, had this correct. Using two values of x to find A and B was by far the most popular method, although some candidates set up and solved simultaneous equations.

For part (b), there are several ways of finding the coefficients C and D , and these were all seen in candidates' work. The anticipated method of long division was seen less often than multiplying through by the denominator and then using two values of x or equating coefficients of powers of x or a mix of the two. Those who used long division sometimes failed to interpret their result. Using x = 0 readily gives the value of D, but those who used $x = \frac{3}{2}$ in finding C often made a mistake; the value of C is most readily found by equating the coefficients of x^3. Those candidates who did not use an efficient method here would have wasted time that may have been more gainfully used in other questions.

A significant minority failed to make any progress in part (c) through not seeing the connection to parts (a) and (b). Most immediately went for an incorrect logarithmic integral and scored no further marks. Those who did attempt to use the results from parts (a) and (b) often made sign and / or coefficient errors during the integration. Some candidates who otherwise had this part all correct, failed to score the last mark because they made an error in attempting to manipulate their logarithms to the required form.

Q5.

In part (a)(i), the partial fractions were found correctly by most students, substituting x = 0 and x = 3 being preferred over setting up simultaneous equations or a mix of both.

Most students went on to integrate the partial fraction form correctly, although some omitted the brackets on In (x – 3) , but this was condoned. Some students were confused by the integral of $\frac{1}{x}$, with some trying to make this become x° . Very few students produced complete nonsense for both integrals, although some differentiated.

Students attempted part (b)(i) by long division or equating coefficients and were usually successful in correctly finding all three coefficients. Those who chose equating coefficients often did far more work than was necessary, as they multiplied out and regrouped the whole expression and were more likely to make an error. Relatively few used $x = -\frac{1}{2}$ to find r first, and those who did rarely progressed beyond a value for r.

In part (b)(ii), although most students attempted to use their result from part (b)(i), most of these failed to realise that r was not multiplied by (2x – 1) , and simply subtracted r from q to give a relatively simple polynomial integral, which gained no credit. Those who did use algebra correctly to set up the integral, usually gave the integral as a log term, but often with an error in the coefficient. Here too, some students differentiated rather than integrated.

Q6.

The majority of candidates did well on this question, with nearly half the candidates scoring 10 of the 12 marks or more.

In part (a) candidates were generally confident and competent in finding partial fractions and A and B were usually found correctly. The most common method was to substitute x = –1 and
x = –3/5. The latter occasionally caused arithmetic problems. However, those who chose to use simultaneous equations or a mix of the two usual methods were just as successful.

In part (b) was pleasing to see that candidates could generally use the binomial expansions correctly and accurately and there were many complete correct solutions. The most common

errors occurred in the expansion of (3 + 5x)$^{-1}$. Instead of $3^{-1}\left(1+\frac{5}{3}x\right)^{-1}$ some used $3\left(1+\frac{5}{3}x\right)^{-1}$.

Some omitted the brackets around $\left(\frac{5x}{3}\right)^2$ too, although some continued correctly from here.
Almost all knew they were to combine the two series to get to the final result, although many sign and arithmetic errors were seen in attempting to do this. Very few misused the

constants A and B found earlier, with just a few raisin g their result to these powers. Few candidates attempted this question by multiplying their expansions together.

In (c) many candidates showed they were not familiar with finding the range of valid values of x. Some just stated $-1 < x < 1$ while others assumed the answer to be $-5/3 < x < 5/3$. Some didn't even use an inequality but gave specific values for which they believed the expansions were not valid. Those who did find the correct limits often included the equality, so lost a mark.

Q7.

In part (a) a repeated term partial fractions expression such as this is the most complex form in the specification, but most candidates approached it with confidence and demonstrated they knew how to find the coefficients. Most chose to use values of x to find C and A in that order, and then either a third value of x, or equated coefficients in order to find a value for B. This approach was generally more successful than those who equated coefficients to set up and solve three simultaneous equations. Most had the value of C correct, and similarly A although

$\frac{1}{4}A = 1$ often resulted in $A = \frac{1}{4}$. However, about 60 % of the candidates had all of A, B and C correct.

In (b) many candidates ignored the partial fractions they had just found, and attempted all sorts of nonsensical ways of "integrating" the right hand side, some before even attempting to separate the variables, using what appeared to be an attempt to use parts. Some who did separate the y term to the left hand side, then attempted to multiply out the right hand side and integrate term by term, taking no notice of denominators. About 40% of the candidates gained no credit in part (b). However, many did know the first step in solving the differential equation was to separate the variables and use the partial fractions. The notation was often poor as was the algebra in handling the 2 $\sqrt{y}$, the latter often appearing as such on the left hand side and not as its reciprocal. Many mistakes were made in attempting to integrate the various terms. Indices caused problems for many with the term $\frac{1}{\sqrt{y}}$ being expressed as both $y^{\frac{1}{2}}$ and $y^{-\frac{1}{2}}$. Credit was given for integrating $\frac{k}{\sqrt{y}}$ to $2k\sqrt{y}$, for any k. In the two log integrals on the right-hand side coefficient and sign errors were common and many thought the third term had a logarithmic integral as well. Others assumed this term integrated to $\frac{x}{1-x}$ as given in the answer. Some candidates did integrate this term correctly by inspection, whereas others used substitution, also correctly. Most candidates included a constant and tried to find it using the boundary conditions given, but they were only credited with the correct value if was exact, and found from a correct expression. The penultimate mark was for correctly combining the logarithm terms in a candidate's solution; many failed to do this correctly, multiplying terms when they should have been dividing or vice versa depending on signs. Those who had made errors and so didn't have all the coefficients equal to 2, did make some valiant efforts using powers to combine their terms, and got the credit if it was done correctly. Everything had to be correct to gain the last mark, and this was achieved by rather less than 5% of the candidates.

Q8.

In part (a), most candidates knew to multiply through by the denominator and many did this correctly and went on to find A and B successfully. However, some candidates did not apparently know how to handle the 2, some omitting it altogether and some including it but not multiplying it by the denominator. Some candidates multiplied out the denominator and divided by the resulting quadratic expression to separate out the 2, then worked with the resulting algebraic fraction. Although this was unnecessary, it was a perfectly acceptable method. Relatively few candidates used the method of setting up simultaneous equations to find A and B.

In part (b), virtually all candidates knew they were to use the partial fractions to do the integration, and most recognised that log integrals were involved. The common error was to miss the multiplier $\frac{1}{5}$ when integrating $(5x - 1)^{-1}$. Several candidates inverted the coefficient, giving it as $\frac{5}{7}$. Although most candidates integrated 2 to $2x$, many omitted the arbitrary constant.

Q9.

(a) (i) The vast majority of candidates found the partial fractions correctly, most substituting the two expected values of x. Some proceeded by setting up and solving simultaneous equations and some a hybrid of both methods, but few errors were made. A few lost all the marks here through having an algebraically incorrect opening line.

(ii) Here too, most candidates knew these were log integrals with relatively few showing no knowledge of this. The common error was to omit the $\frac{1}{3}$ and a few candidates didn't have their log expressions in brackets and were penalised.

(b) Various attempts at this question were seen, ranging from no attempt to a time consuming approach involving setting up simultaneous equations in P, Q and R following substitution of values of x, usually 0, 1 and 2. Most candidates who attempted this either set up just two equations, and thus found it was insoluble, or simply made errors in the algebra. The most successful and efficient method was algebraic long division. Relatively few candidates showed any knowledge of the technique of manipulating the numerator, as anticipated in the mark scheme. Some attempted to factorise either the numerator or denominator, or both, and made no further progress because these expressions did not factorise.

Q10.

Part (a) was generally done very well with most candidates gaining full marks, although many methods were seen in the finding of A and B, ranging from inspection to division to rearranging the numerator. Most candidates also integrated correctly, although many omitted the constant of integration and were penalised.

In part (b), most candidates had the opening line of clearing fractions correct, although a few had an extra $(5 - x)$ and could score no further marks. The finding of P, Q and R was done accurately and concisely by many candidates. Such candidates had usually substituted 5, $-\frac{1}{3}$ and 0 for x.

The occasionally-seen error was in finding the value of Q. Those candidates who proceeded by multiplying out and equating coefficients often made numerical and/or algebraic errors and sometimes abandoned their efforts when they saw the unlikely looking numbers emerging. This approach cost them marks and time. Some candidates used a hybrid of both methods.

All candidates who attempted part (b)(ii) knew that they were to use the partial fractions and, with the occasional exception, knew that the first two integrals were logarithmic. However, many mistakes were made in the coefficient's value and/or sign. Relatively few were successful on the third integral. Many just had the sign incorrect, although some assumed it too was a logarithm, and some expanded $(5 - x)^2$ into a quadratic expression and ended up going nowhere with it.

Q11.

Many candidates were successful in this question, particularly those who used long division which was the most efficient way to solve the problem. Those candidates who chose a method based on equating coefficients often made algebraic errors. Some candidates wrote $f(x) = (ax^2 + bx + c)(3x - 1) + c$ which led to confusion and few proved the first constant c to be zero.

Q12.

The unsimplified form of the binomial expansion in part (a) was usually correct. However in attempting to simpifly their expressions many candidates made sign or coefficient errors, with answers such as $1 - x + x^2$ or $1 + x + 2x^2$. There were a pleasing number of correct expansions however. In part (b)(i) virtually all candidates showed they knew how to find partial fractions. Those who substituted $x = 1$ and $x = 2/3$ usually found the values of A and B correctly, with a few sign errors seen. Substitution of other values of x, or using simultaneous equations tended to be less successful.

There were many complete correct solutions to part (b)(ii) and most candidates demonstrated they knew in principle how to find the expansion. Many, though, made a mistake somewhere in their working. The most common error lay in the expansion of $(2 - 3x)^{-1}$. Many forgot to put 2 to the power -1 so the expansion of $(1 - 3x/2)^{-1}$ was often multiplied by 6 instead of 3/2. Also some expanded $(1 - 3x)^{-1}$ instead of $(1 - 3x/2)^{-1}$. Some candidates forgot to multiply their separate expansions by their values of A and B or to add them together to complete the expansion.

The negative sign in $A = -2$ was often overlooked when combining the two expansions. Some candidates used the method of finding the two expansions and then finding the product of $(3x - 1)$ and two binomial expansions. This nearly always led to errors. Candidates should realise that there is often a link between parts of a question. The use of the partial fractions led to a more sensible method with far less algebra involved.

The majority of candidates scored no marks in part (c). Answers such as x > 2/3, 1 < x < 2/3 abounded and there was confusion between 2/3 and 3/2. Many candidates apparently just made a guess, with x not equal to 1 or 2/3 being a common one. Many gave an inequality in which a modulus was less than a negative number.

Q13.

This question, surprisingly, was not done well. Part (a) was intended as a straight forward partial fractions question, although some candidates did not see it that way. Many apparently guessed at the value of k, 3 being a common wrong answer. Some candidates replaced the k with the conventional A and B, and then demonstrated that A = B. Many of these, having arrived correctly at 6A = 3, then deduced that A = 2. This 2, or other values, often became $\frac{1}{2}$ in part (b) without any supporting evidence and so lost further marks. Very few candidates in this situation seemed to have checked for a mistake, despite the $\frac{1}{2}$ being given in part (b).

In part (b), most candidates picked up a mark for recognising a log integral, although the coefficients were often incorrect, particularly the sign in integrating the $\frac{1}{3-x}$ term. However, most candidates could evaluate their log expression correctly using the laws of logs.

Q14.

Most candidates scored well on this question although there were some unnecessarily long solutions.

In part (a)(i), most candidates substituted $x = \frac{1}{2}$, correctly into the polynomial, but many lost a mark by not demonstrating that the result came to zero by showing the required arithmetic and/or by not interpreting the result as demonstrating that (2x – 1) is a factor. For part (a)(ii), many candidates did an algebraic long division or multiplied out the given expression and equated the coefficients. Both techniques lead to the correct answer, but the value of q could be written down by inspection and the value of p should follow in a couple of lines of working. This is where many candidates spent an unnecessarily long time in their working. Most candidates answered part (a)(iii) correctly, although, surprisingly, the common error was to just omit the 4 when factorising the numerator.

Part (b) was attempted via a variety of methods. The most concise of these was to divide $2x^2$ by $x^2 + 2x - 15$ and interpret the remainder. Many candidates who took the more conventional partial fractions approach had not realised that A = 2 was immediate and so they set up simultaneous equations in three unknowns, often making a mistake in their solution.

Those who substituted suitable values of x were usually more successful. Some candidates wrote the right-hand side as $\frac{B}{x+5} + \frac{C}{x-3}$ with some then going on to find the values of B and C correctly and recombining their fractions to get the required form. However, this is an unnecessarily long route to the solution, and mistakes were often made along the way, not least in the opening line when candidates were attempting to multiply through by the denominator.

Q15.

Nearly all candidates successfully found the partial fractions in part (a), most substituting $x = 1$ and $x = –1$; although some used other values or equated coefficients.

In part (b) most candidates then went on to use their partial fractions to integrate the given expression correctly. The error that was sometimes seen was to integrate $(x – 1)^{–1}$ to $(x – 1)^0$, and similarly for $(x + 1)^{–1}$.

Most candidates could make a start to part (c) of the question, but many rearranged the 2 and 3, and this detracted from the realisation that with y and 3 taken to the other side of the equation, they had the integral they had just done in part (b). Some who did realise this, proceeded to give an incorrect integral because they had "lost" the 2. Most candidates knew they needed to separate the variables but for many candidates the algebra and ensuing attempts at integration were full of errors. Turning the denominators y and $(x^2 – 1)$ into numerators was common, as was integrating $(x^2 – 1)^{–1}$ to an expression involving ln $(x^2 – 1)$.

Those who had taken the 2 to the left hand side, often got confused with the integral of $(2y)^{–1}$, this becoming ln $2y$. The problem for most candidates seemed to be that they associated the 2 in the question with the 2 in the answer, although the 2 in the answer came from the given values of x and y. Most candidates did include an arbitrary constant and tried to use the given values to find it, often claiming, after some dubious manipulation, that they had derived the given result. Few fully correct answers were seen, but where they were, it was reassuring to see confident use of the log laws in deriving the result.

Q16.

This question was generally done well with many candidates gaining full marks. Marks tended to be lost in part (a) rather than part (b), possibly because the latter was a more conventional partial fractions problem. There were several different approaches to part (a) including manipulating the numerator to contain $A(x – 3)$, algebraic division and cross multiplying and using partial fraction type techniques, or some mix of these. Errors tended to be algebraic or numerical. In part (a)(ii) it was notable that some candidates did not attempt a logarithmic integral, although they went on to do so in part (b).

In part (b) virtually all candidates scored well, either using the expected two values of x or equating coefficients to find the values of P and Q. Again a few made algebraic or numerical errors, with very few cross multiplying incorrectly. Similarly virtually all candidates went on to give some form of logarithmic integrals, sometimes with errors in the coefficients. Very few non-sensible attempts at the integral were seen.

Q17.

This question was generally done very well apart from part (c)(ii).

In part (a), most candidates did the two expansions accurately and clearly, with the common error being not to raise the 3 to the same power as x.

For part (b), most candidates knew what was required and used an appropriate technique, either substituting values of x or setting up simultaneous equations. A few candidates added in a constant, which was acceptable if they showed it was zero. Errors tended to be algebraic and/or numerical.

For part (c)(i), most candidates knew that they were supposed to use their result from part (b), although some did not indicate this and started with an incorrect expression, having inverted their coefficients, and thus scored no marks. Another error often seen here was to raise $(1 + x)$ and $(1 + 3x)$ to the power of the coefficients found in part (b).A few candidates tried to use their expansions in place of the denominator. An algebraic slip in simplification cost many otherwise successful candidates the final mark. A few candidates did not use the partial fractions and wrote the expression as a product using their expansions; this is quite acceptable but involves a lot more work to achieve the correct answer.

In part (c)(ii), most candidates seemed unfamiliar with defining a valid range for a binomial expansion. Some attempted to use the modulus notation, but errors such as $|x| < -3$ showed a lack of understanding of it. Some more successful candidates only gave half the range, omitting the negative numbers, and the inequality often went the wrong way.

Many candidates argued that $x \neq -1$ and $x \neq -\frac{1}{3}$ defined the range as you cannot divide by zero.

Q18.

This question was usually done very well, with many candidates scoring highly. Binomial expansions and use of partial fractions were seen to be generally well known with most of the algebraic manipulation being done accurately. Most knew the result for part (a) (i) although some made errors with the sign of the x term and lost both marks. In part (ii) most made the expected start of $\frac{1}{3}(1 - \frac{2}{3}x)^{-1}$ and then gave a detailed expansion rather than use the result from part (i) with $\frac{2}{3}x$ replacing x in their expansion. Some candidates had 3 where $\frac{1}{3}$ should be, it just 'becoming' $\frac{1}{3}$ to get the given answer. A few candidates manipulated powers of 2 and 3 for no good reason other than to achieve the given answer, most unconvincingly. Some used the expansion of $(a + bx)^n$ from the formula book, often successfully but the few who attempted to use $\binom{n}{r}$ with $n = -1$ gained no credit. The response to part (b) was similar to that in part (a) (i).

Most candidates knew how to start part (c) although the accuracy of the algebraic expression written down often left something to be desired. Those who substituted $x = 1$, $x = \frac{3}{2}$ and $x = 0$ were usually more successful in getting A, B and C correctly than those who multiplied out and set up simultaneous equations by equating coefficients. Solutions to these equations were often just written down with no working seen. This gained full credit if they were correct, but usually none otherwise. Most attempted part (d) using the results from previous parts of the question. Many did this correctly although some made algebraic or numerical slips and some candidates used the reciprocals of A, B and C. A few did not use their partial fraction expression but multiplied out the product of the series directly often resulting in an algebraic error.

Q19.

There were many good answers to this question, with full or nearly full marks.

Part (a) Partial fractions is an area that is generally known well, although the presence of the 3 led some candidates to make an error, some ignoring it altogether, and others not multiplying it by the common denominator. Some candidates simplified the given expression by dividing it out, which was perfectly valid. Those candidates who substituted x = 1 and x = – usually found the coefficients correctly; many candidates who chose to set up simultaneous equations by equating coefficients made an error.

Part (b) Virtually all candidates knew they were to use the partial fractions to do the integral and that it involved ln(3x – l) and ln(x – l), and very few nonsense "integrations" were seen. The common errors were in the coefficients with the 1/3 from 3xbeing missed, or the coefficients given as fractions 1/6 and 1/4.

Q20.

Part (a) This was done very well with most candidates showing confidence and accuracy with the partial fractions. Both techniques of substituting suitable values of x or determining unknown coefficients using simultaneous equations were seen. Any errors made were usually arithmetical.

Part (b) Most candidates knew that they were to use the partial fractions in attempting the integral, and most knew these were 'ln' integrals, although some attempts at independently integrating numerator and denominator were seen. The common error in the ln integrals was the omission of the coefficient 1/2.